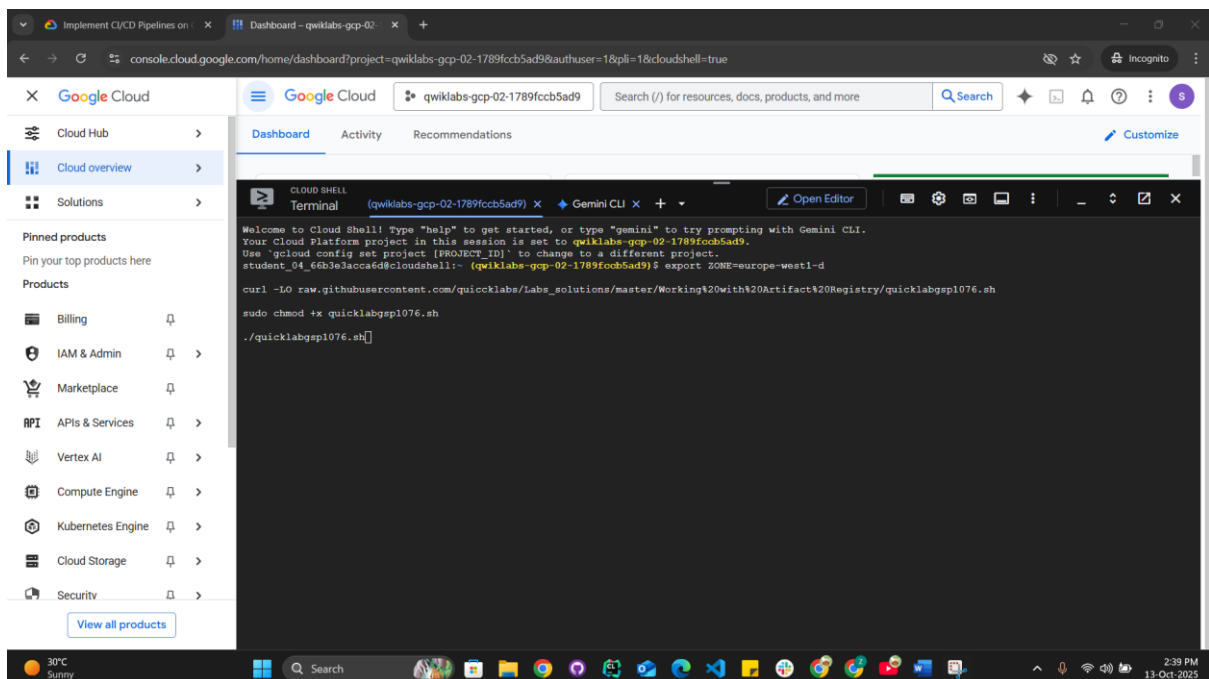
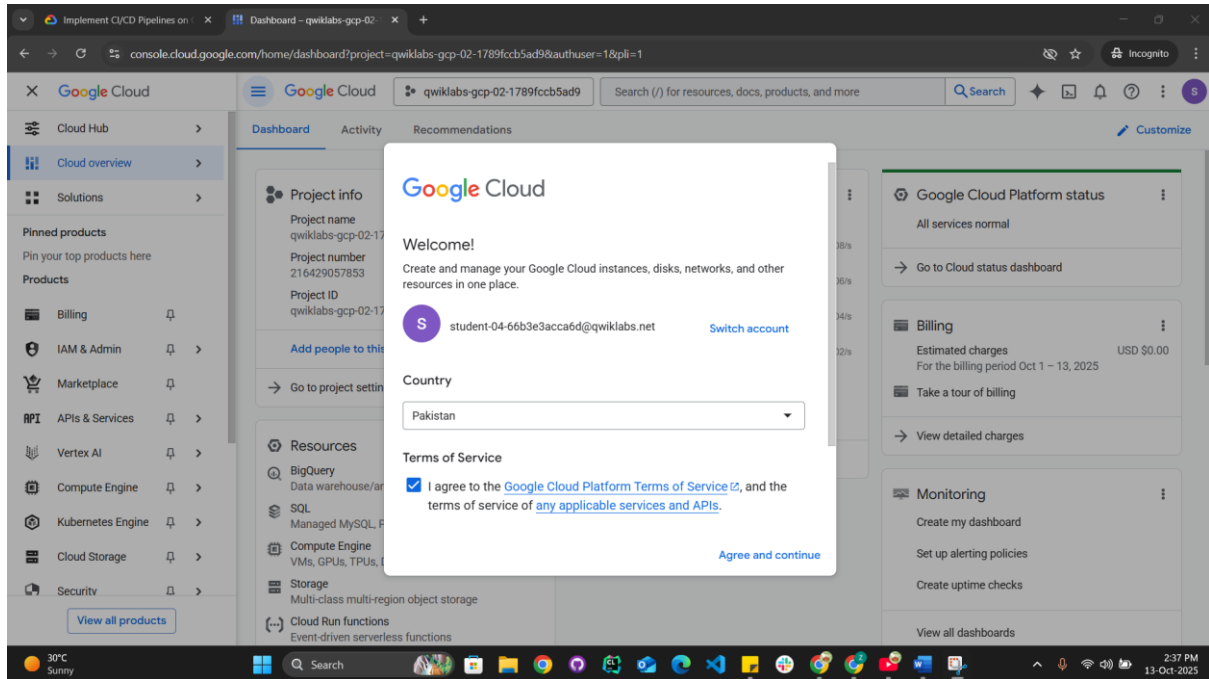
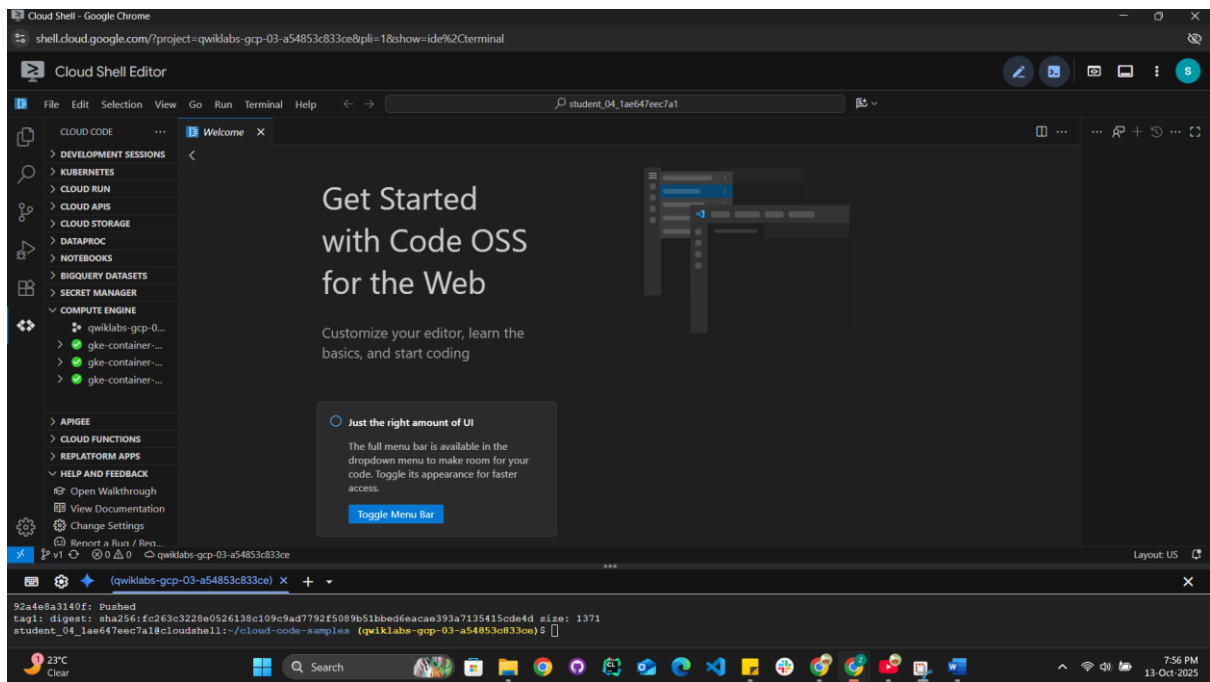
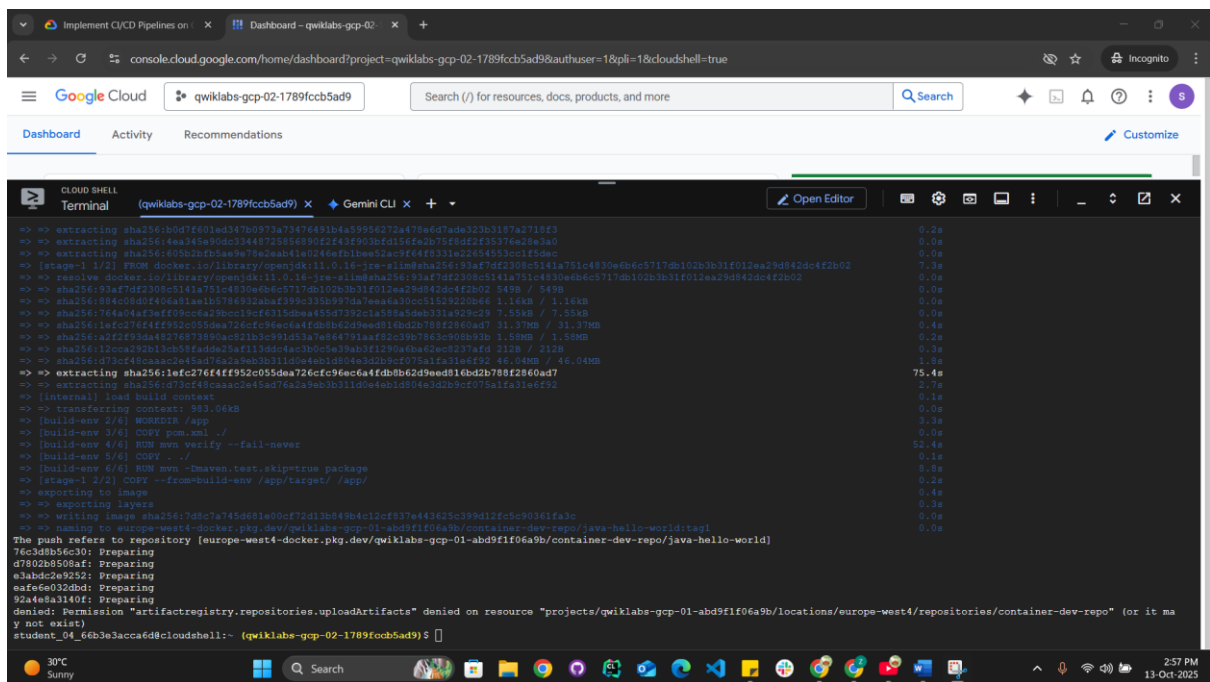
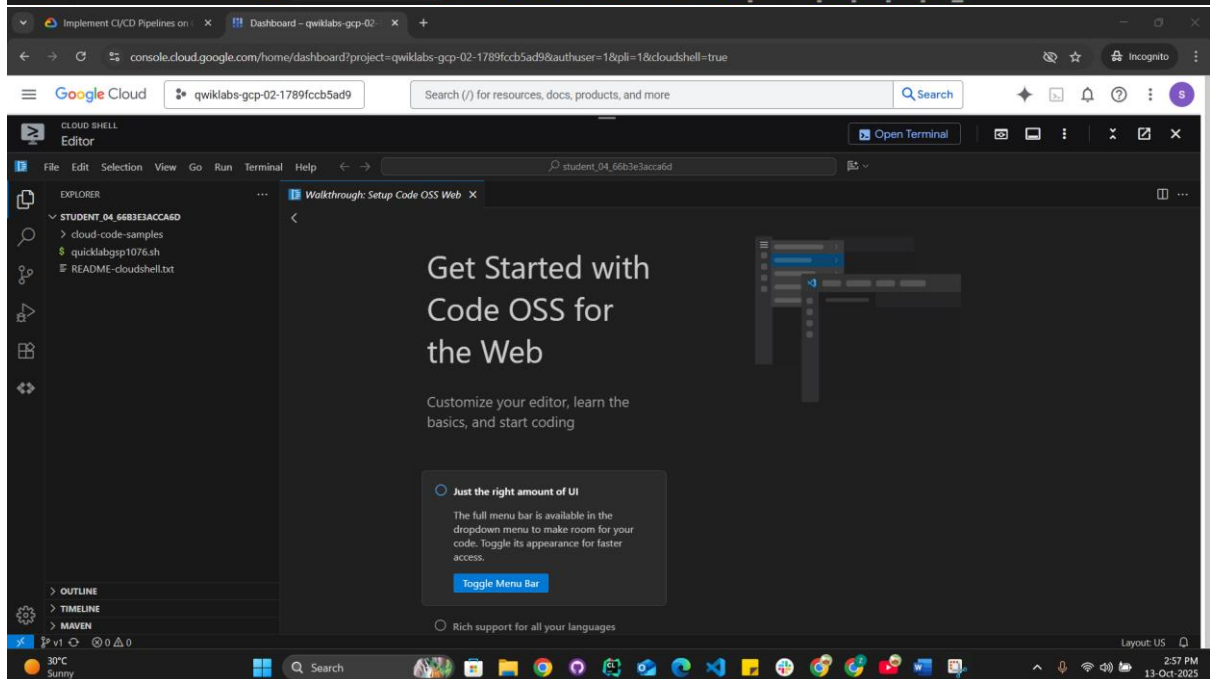
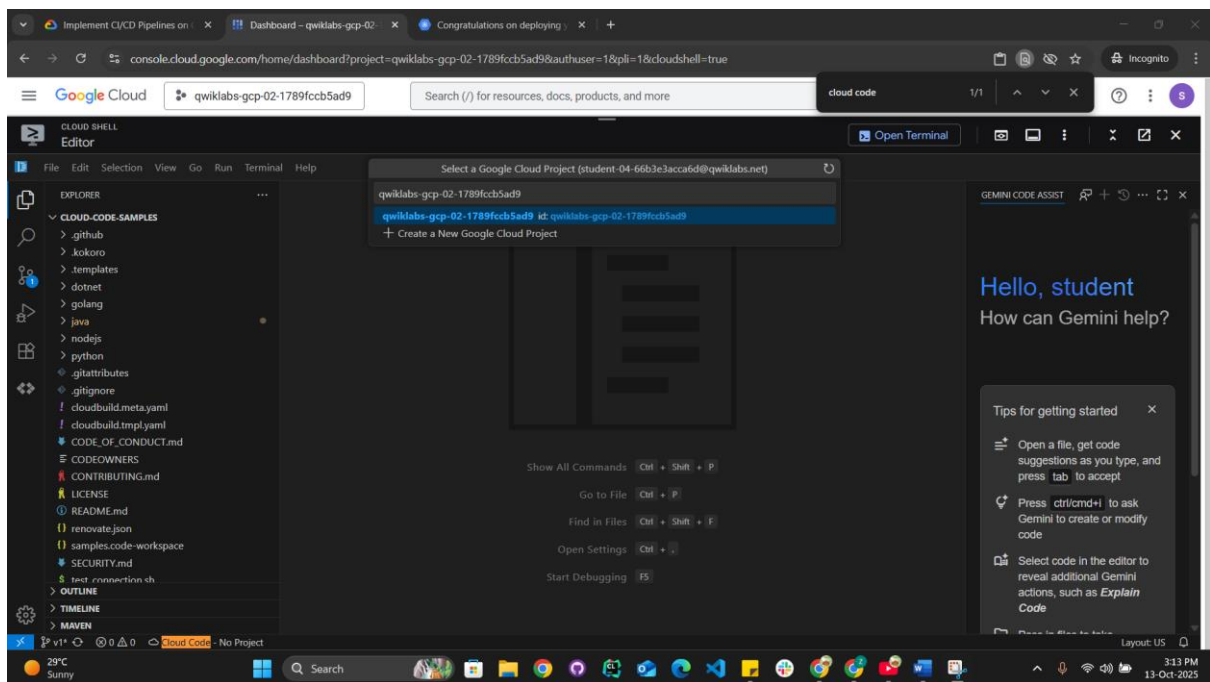
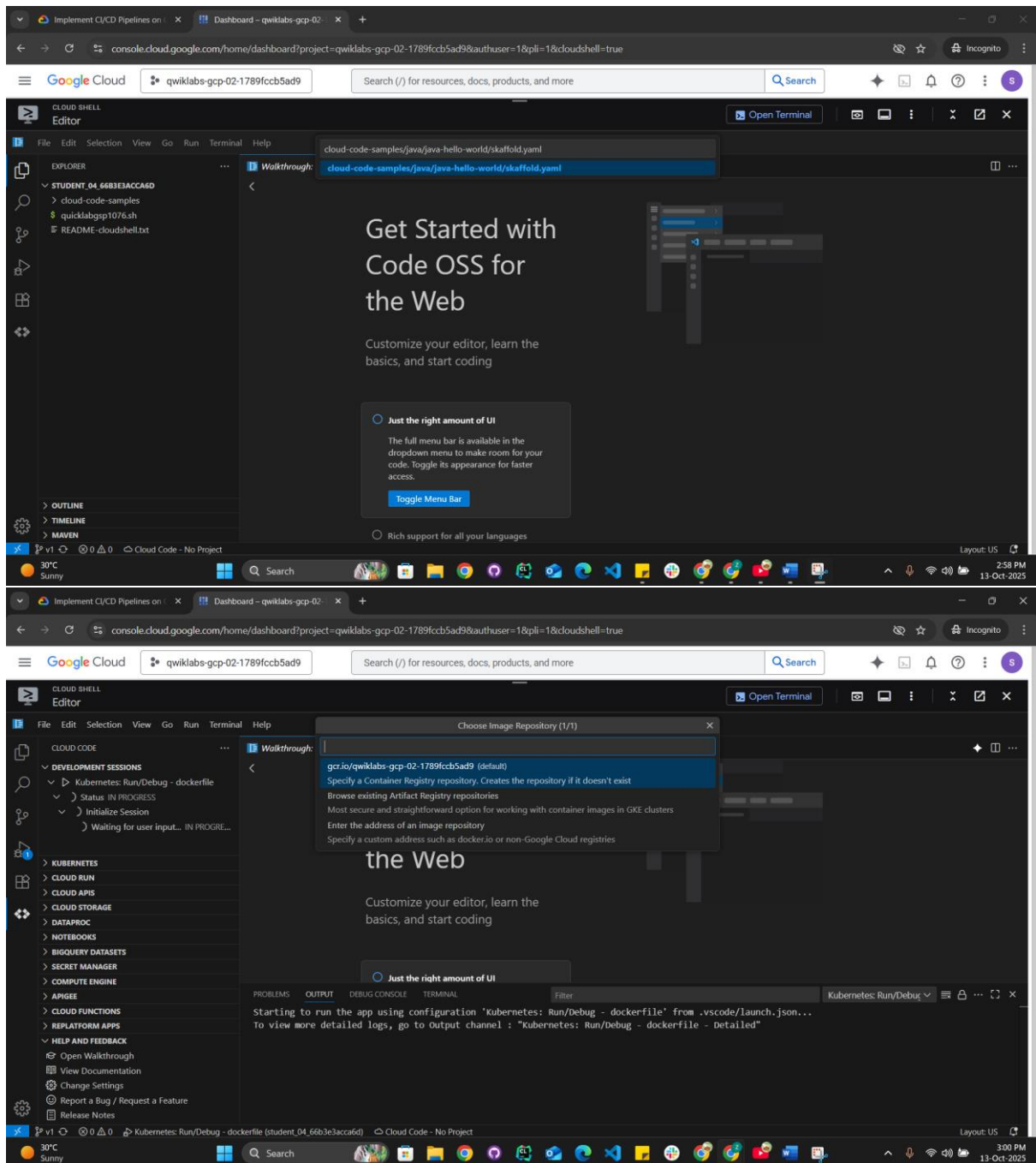


1. Working with Artifact Registry









Implement CI/CD Pipelines on ...Dashboard - qwiklabs-gcp-02Loading...console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-02-1789fcb5ad9&authuser=1&pli=1&cloudshell=true

Google Cloudqwiklabs-gcp-02-1789fcb5ad9Search (/) for resources, docs, products, and moreSearch

CLOUD SHELLEditor

FileEditSelectionViewGoRunTerminalHelpstudent_04_66b3e3acca6dOpen Terminal

CLOUD CODE

DEVELOPMENT SESSIONS

KUBERNETES

CLOUD RUN

CLOUD APIS

CLOUD STORAGE

DATAPROC

NOTEBOOKS

BIGQUERY DATASETS

SECRET MANAGER

COMPUTE ENGINE

APIGEE

CLOUD FUNCTIONS

REPLATFORM APPS

HELP AND FEEDBACK

Open Walkthrough

View Documentation

Change Settings

Report a Bug / Request a Feature

Release Notes

Walkthrough: Setup Code OSS Web

Get Started with Code OSS for the Web

Customize your editor, learn the basics, and start coding

Just the right amount of UI

PROBLEMSOUTPUTDEBUG CONSOLETERMINALFilterKubernetes: Run/Debug

Resource deployment/java-hello-world status completed successfully
Status check succeeded
*****URLS*****
Deploy completed
Forwarded URL from service java-hello-world-external: http://localhost:4503
Update succeeded

v1000Kubernetes: Run/Debug - dockerfile (student_04_66b3e3acca6d)Cloud Code - No Project

Deployment SuccessfulLayout: US

30°C Sunny

Search

3:03 PM13-Oct-2025

Implement CI/CD Pipelines on ...Dashboard - qwiklabs-gcp-02Congratulations on deploying ...4503-cs-111092e2-f7f8-49bf-887b-4ebd9fb0d0fb.qf-asia-southeast1-wlry.cloudshell.dev

It's running!

Congratulations, you successfully deployed a Kubernetes application with Cloud Code!

What's next?

Learn more about working with Kubernetes in Cloud Code for [VSCode](#) or [IntelliJ](#)

Get started with [Google Cloud APIs](#) in Cloud Code

Watch how Cloud Code can optimize your [GKE Developer Workflow](#)

Explore more options with

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Search

3:03 PM13-Oct-2025

Implement CI/CD Pipelines on > Dashboard - qwiklabs-gcp-02 > Congratulations on deploying > +

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-02-1789fcb5ad9&authuser=1&pli=1&cloudshell=true

Google Cloud qwiklabs-gcp-02-1789fcb5ad9 Search (/) for resources, docs, products, and more

CLOUD SHELL Editor

File Edit Selection View Go Run Terminal Help

student_04_66b3e3acca6d

Walkthrough: Setup Code OSS Web Hello

cloud-code-samples > java > java-hello-world > src > main > java > cloudcode > helloworld > web > HelloWorldController.java

```
1 package cloudcode.helloworld.web;
2
3
4 import org.springframework.stereotype.Controller;
5 import org.springframework.ui.Model;
6 import org.springframework.web.bind.annotation.GetMapping;
7
8 /**
9  * Defines a controller to handle HTTP requests.
10 */
11 @Controller
12 public final class HelloWorldController {
13
14     /**
15      * Create an endpoint for the landing page
16      * @return the index view template with a simple message
17     */
18     @GetMapping("/")
19     public String helloworld(Model model) {
20         String message = "It's updated!";
21         model.addAttribute("message", message);
22         return "index";
23     }
24 }
25
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL Filter

Kubernetes: Run/Debug

Deploy completed

Forwarded URL from service java-hello-world-external: <http://localhost:4503>

Update succeeded

Watching for changes...

To disable watch mode for subsequent runs, set watch to false in your launch configuration /home/student_04_66b3e3acca6d/.vscode/launch.json and relaunch the application.

Update initiated

Do you want to install the recommended 'Extension Pack for Java' extension from vsjava for the Java language? [Install](#) [Show Recommendations](#)

30°C Sunny

304 PM 13-Oct-2025

Implement CI/CD Pipelines on... Dashboard - qwiklabs-gcp-02... Congratulations on deploying... 4503-cs-11109262-f7f8-49bf-887b-4ebd9b0d0fb.q1-asia-southeast1-wly.cloudshell.dev

It's updated!

Congratulations, you successfully deployed a Kubernetes application with Cloud Code!

What's next?

Learn more about working with Kubernetes in Cloud Code for [VSCode](#) or [IntelliJ](#)

Get started with [Google Cloud APIs](#) in Cloud Code

Watch how Cloud Code can optimize your [GKE Developer Workflow](#)

Explore more options with

30°C Sunny

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-02-1789fccb5ad9&authuser=1&pli=1&cloudshell=true

Google Cloud qwiklabs-gcp-02-1789fccb5ad9 Search (/) for resources, docs, products, and more

CLOUD SHELL Terminal (quiklabs-gcp-02-1789fccb5ad9) Gemini CLI + Open Editor

```
>> >> extracting sha256:80b2b3f3a9e78e1e04e6f1b2e82c0f44f8331e225848530c1f5d8c 0.0s
>> [stage-1 1/2] RUN docker.io/library/openjdk:11.0.16-jre-slim&sha256:93af7df2200c5141a751c4830e4bdc5717db102b3b31f012ea29d4842d04f2b02 7.3s
>> resolve docker.io/library/openjdk:11.0.16-jre-slim&sha256:93af7df2200c5141a751c4830e4bdc5717db102b3b31f012ea29d4842d04f2b02 0.0s
>> sha256:93af7df2200c5141a751c4830e4bdc5717db102b3b31f012ea29d4842d04f2b02 549B / 549B 0.0s
>> sha256:80b2b3f3a9e78e1e04e6f1b2e82c0f44f8331e225848530c1f5d8c 1.16MB / 1.16MB 0.0s
>> sha256:76a044f3eef0b0ca29b0c150e485d378c1a33845d0031a928c09 7.55MB / 7.55MB 0.0s
>> sha256:1efc276f4ff952c055dea726fc96ec64fdb8b61d9eed16bd2b788f22860ad7 31.37MB / 31.37MB 0.4s
>> sha256:a22f95da48276873890ac821b3c991d51a7e864791aaf92c39b7863c90bb93b 1.58MB / 1.58MB 0.2s
>> sha256:12c0e49a1b12b284d0a5a1110b4f6c0c0c49ad3f129040e0e0c227e48 212B / 212B 0.0s
>> sha256:d73cf40caac0e45ad76a2a5eb3b311d0e4eb1d80e3d2b5cf075a1fa31e6f92 46.04MB / 46.04MB 1.6s
>> extracting sha256:1efc276f4ff952c055dea726fc96ec64fdb8b61d9eed16bd2b788f22860ad7 75.4s
>> extracting sha256:d73cf40caac0e45ad76a2a5eb3b311d0e4eb1d80e3d2b5cf075a1fa31e6f92 2.7s
>> [itwenal] /usr build context 2.1s
>> transferring context: 851.04kB 0.0s
>> [build-env 2/6] WORKDIR /app 3.3s
>> [build-env 3/6] COPY pom.xml ./ 0.0s
>> [build-env 4/6] RUN mvn verify --fail-never 52.4s
>> [build-env 5/6] COPY ./ 0.1s
>> [build-env 6/6] RUN mvn -Dmaven.test.skip=true package 8.8s
>> [stage-1 2/2] COPY --from=build-env /app/target/ /app/ 0.2s
>> exporting to image 0.4s
>> exporting layers 0.3s
>> writing image sha256:7d8c7a745d4e81e00c72d12b849b4c12c837e443625c399d12fcb90361fa3c 0.0s
>> pushing to europe-west4-docker.pkg.dev/quiklabs-gcp-01-abd9f1f06a9b/container-dev-repo/java-hello-world:tag1 0.0s
The push refers to repository (europe-west4-docker.pkg.dev/quiklabs-gcp-01-abd9f1f06a9b/container-dev-repo/java-hello-world:tag1)
76c3d9b56c30: Preparing
d7802b508af: Preparing
e1abd2e9252: Preparing
eafe60328bd: Preparing
92a4e8a3140f: Preparing
denied: Permission "artifactregistry.repositories.uploadArtifacts" denied on resource "projects/quiklabs-gcp-01-abd9f1f06a9b/locations/europe-west4/repositories/container-dev-repo" (or it may not exist)
student_04_6b3e3acca6@cloudshell:~ (quiklabs-gcp-02-1789fccb5ad9) $ gcloud artifacts repositories create container-dev-java-repo \
--repository-format=maven \
--location=europe-west1 \
--description="Java package repository for Container Dev Workshop"
Create request issued for: [container-dev-java-repo]
Waiting for operation [projects/quiklabs-gcp-02-1789fccb5ad9/locations/europe-west1/operations/4242fa00-49c6-461d-9370-062aead5611] to complete....done.
Created repository [container-dev-java-repo].
student_04_6b3e3acca6@cloudshell:~ (quiklabs-gcp-02-1789fccb5ad9) $
```

30°C Sunny

3:07 PM 13-Oct-2025

Software Development & Cloud | Implement CI/CD Pipelines on Google Cloud

cloudskillsboost.google/paths/76/course_templates/691/labs/594611

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← Main menu

10 Implement CI/CD Pipelines on Google Cloud

Course - 1 Complete hour 15 minutes

Course overview

- Implement CI/CD Pipelines on Google Cloud
- Working with Artifact Registry
- Google Kubernetes Engine Pipeline using Cloud Build
- Continuous Delivery with Google Cloud Deploy

DevSecOps > Course > Implement CI/CD Pipelines on Google Cloud >

End Lab 00:30:29

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

[Open Console](#)

Project ID: qwiklabs-gcp-03-a54853cl

Username: student-04-1ae647eec7a1l

Password: iLv3f3sXh91S

3. In the Cloud console go to **Artifact Registry > Repositories** and notice your newly created Maven repository named `container-dev-java-repo`, if you click on it you can see that it's empty at the moment.

Click **Check my progress** to verify the objective.

Create a Java package repository

[Check my progress](#)

Set up authentication to Artifact Repository

- Use the following command to update the well-known location for Application Default Credentials (ADC) with your user account credentials so that the Artifact Registry credential helper can authenticate using them when connecting with

Lab instructions and tasks 100/100

GSP1076

Overview

Objectives

Setup and requirement

Task 1. Prepare the lab environment

Task 2. Working with container images

Task 3. Integration with Cloud Code

Task 4. Working with language packages

Check complete. Points earned: 20. Message: Assessment Completed!

[Next >](#)

Software Development & Cloud | Implement CI/CD Pipelines on Google Cloud

cloudskillsboost.google/paths/76/course_templates/691/labs/594611

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10 Implement CI/CD Pipelines on Google Cloud

Course - 1 Complete hour 15 minutes

Course overview

- Implement CI/CD Pipelines on Google Cloud
- Working with Artifact Registry
- Google Kubernetes Engine Pipeline using Cloud Build
- Continuous Delivery with Google Cloud Deploy

DevSecOps > Course > Implement CI/CD Pipelines on Google Cloud >

End Lab 00:30:04

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

[Open Console](#)

Project ID: qwiklabs-gcp-03-a54853cl

Username: student-04-1ae647eec7a1l

Password: iLv3f3sXh91S

3. In the Cloud console go to **Artifact Registry > Repositories** and notice your newly created Maven repository named `container-dev-java-repo`, if you click on it you can see that it's empty at the moment.

Click **Check my progress** to verify the objective.

Create a Java package repository

[Check my progress](#)

Set up authentication to Artifact Repository

- Use the following command to update the well-known location for Application Default Credentials (ADC) with your user account credentials so that the Artifact Registry credential helper can authenticate using them when connecting with

Checkpoints

- Enable Google Services and create container cluster [Check my progress](#) 40 / 40
- Working with container images [Check my progress](#) 20 / 20
- Integration with Cloud Code [Check my progress](#) 20 / 20

Lab instructions and tasks

GSP1076

Overview

Objectives

Setup and requirement

Task 1. Prepare the lab environment

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Task 3. Integration with Cloud Code

Task 4. Working with language packages

[Next >](#)

Software Development & Con...Implement CI/CD Pipelines on...cloudskillsboost.google/paths/76/course_templates/691/labs/594611

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image1/29Z

Apply your skills in Google Cloud consoleGet started

← Main menu

10Implement CI/CD Pipelines on Google Cloud

Course - 1Completehour 15 minutes

Course overview

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

End Lab00:29:54

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Console

Project IDqw1klabs-gcp-03-a54853cl

Usernamestudent-04-1ae647eec7a1l

PasswordiLv93fsXh9IS

Previous

3. In the Cloud console go to **Artifact Registry**. On the left sidebar, click **Repositories** and notice your newly created Maven repository named **cd**. On it you can see that it's empty at the moment.

Click **Check my progress** to verify the objective.

Create a Java package repository

Check my progress

Set up authentication to Artifact Repository

Use the following command to update the well-known location for Application Default Credentials (ADC) with your user account credentials so that the Artifact Registry credential helper can authenticate using them when connecting with

Working with container images20 / 20

Integration with Cloud Code20 / 20

Create a Java package repository20 / 20

environment

Task 2. Working with container images

Task 3. Integration with Cloud Code

Task 4. Working with language packages

Next

23°C Clear

Search

8:04 PM13-Oct-2025

2. Google Kubernetes Engine Pipeline using Cloud Build

```
Cloud Shell - Google Chrome
shell.cloud.google.com/?project=qwiklabs-gcp-03-a83160aaeca9&pli=1&show=ide%2Cterminal

Cloud Shell Editor

Walkthrough: Setup Code OSS Web

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to qwiklabs-gcp-03-a83160aaeca9.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
student_00_21b18e955805@cloudshell:~ (qwiklabs-gcp-03-a83160aaeca9)$ curl -LO https://raw.githubusercontent.com/Itsabhishek7py/GoogleCloudSkillsboost/refs/heads/main/Google%20Kubernetes%20Engine%20Pipeline%20using%20Cloud%20Build/abhishek.sh
sudo chmod +x abhishek.sh
./abhishek.sh
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100  5759  100  5759    0     0  15966      0 --:--:-- --:--:-- --:--:-- 15997

Welcome to Dr Abhishek's Cloud Tutorials! ☁️
Subscribe to the channel: https://www.youtube.com/@drabhishek.5460/videos

Lo Done! [[]]
Your active configuration is: [cloudshell-16715]
Updated property [compute/region].
Operation "operations/acat.p2-880309350907-648fe7f3-e900-4e56-969d-333b958e22e2" finished successfully.
Create request issued for: [my-repository]
Waiting for operation [projects/qwiklabs-gcp-03-a83160aaeca9/locations/us-west1/operations/745a8d57-8e38-45ba-bc41-d363fdb486f0] to complete...done.
Created repository [my-repository].
Note: Your Pod address range (`--cluster-ipv4-cidr`) can accommodate at most 1008 node(s).
Creating cluster hello-cloudbuild in us-west1... Cluster is being configured...working
```

```
Cloud Shell - Google Chrome
shell.cloud.google.com/?project=qwiklabs-gcp-03-a83160aaeca9&pli=1&show=ide%2Cterminal

Cloud Shell Editor

Walkthrough: Setup Code OSS Web x

Extracting ~/Downloads/webi/gh/2.81.0/gh_2.81.0_linux_amd64.tar.gz
Installing to ~/.local/opt/gh-v2.81.0/bin/gh

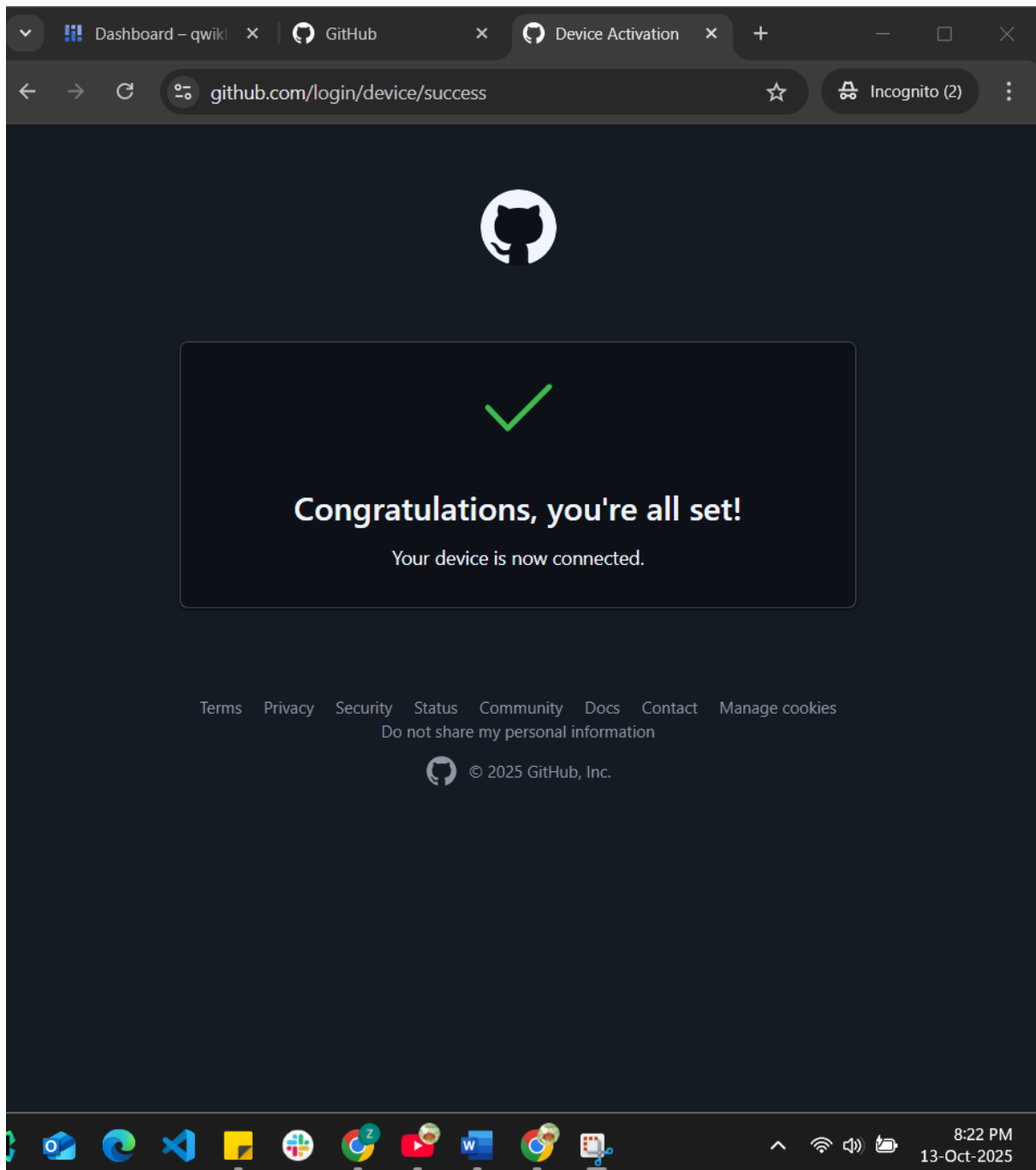
Installed 'gh v2.81.0' as ~/.local/bin/gh

Edit ~/.config/envman/PATH.env to add:
~/.local/bin

>>> ACTION REQUIRED <<<
Copy, paste & run the following command:
source ~/.config/envman/PATH.env
(newly opened terminal windows will update automatically)

? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this host? HTTPS
? Authenticate Git with your GitHub credentials? Yes
? How would you like to authenticate GitHub CLI? Login with a web browser

! First copy your one-time code: 8F5B-658E
Press Enter to open https://github.com/login/device in your browser...
https://github.com/login/device
✓ Authentication complete.
- gh config set -h github.com git_protocol https
✓ Configured git protocol
! Authentication credentials saved in plain text
✓ Logged in as BSSE23058
BSSE23058
BSSE23058
student-00-21b18e955805@qwiklabs.net
GraphQL: Name already exists on this account (createRepository)
GraphQL: Name already exists on this account (createRepository)
Copying gs://splsgsp1077/gke-gitops-tutorial-cloudbuild/CONTRIBUTING.md to file://hello-cloud
build-app/CONTRIBUTING.md
```




```
Cloud Shell - Google Chrome
shell.cloud.google.com/?project=qwiklabs-gcp-03-a83160aaeca9&pli=1&show=ide%2Cterminal

Cloud Shell Editor

Walkthrough: Setup Code OSS Web X

hint: See the 'Note about fast-forwards' in 'git push --help' for details.
# github.com:22 SSH-2.0-1a4a900
[master d39abal] Adding known_host file.
1 file changed, 1 insertion(+)
create mode 100755 known_hosts.github
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 765 bytes | 765.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/BSSE23058/hello-cloudbuild-app
cceb290..d39abal master -> master
% Total      % Received % Xferd  Average Speed   Time    Time     Time  Current
   0      0     0     0    0     0      0      0  --:--:-- --:--:-- --:--:--    0
100 3352 100 3352    0    0 47676    0  --:--:-- --:--:-- --:--:-- 47676
[master 1db9b26] Trigger CD pipeline
1 file changed, 76 insertions(+)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.60 KiB | 546.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/BSSE23058/hello-cloudbuild-app
d39abal..1db9b26 master -> master
✅ Lab is now Completed!
👤 Thanks for using Dr Abhishek's Cloud Tutorials!
👉 Subscribe here: https://www.youtube.com/@drabhishek.5460/videos
student_00_21b18e955805@cloudshell:~ (qwiklabs-gcp-03-a83160aaeca9) $
```

Create trigger – Cl...GitHubDevice Activation +

console.cloud.google.com/cloud-build/triggers;region=u...Incognito (2)

Google Cloud

Cloud Build / Triggers

Create trigger

Repository service

☒ Cloud Build repository service

☐ Developer Connect

Repository generation

☒ 1st gen

☐ 2nd gen

Repository *

Filter Type to filter

No repositories in region

☐ Invert Regex

Show included and ignored

Configuration

Type

☒ Cloud Build configuration

Connect repository

Region: us-west1

☒ Selected

Filter Type to filter

1 of 4 selected

☐ BSSE23058/DBMS-Group-5

☐ BSSE23058/Group-B_SE_Project

☒ BSSE23058/hello-cloudbuild-app

☐ BSSE23058/hello-cloudbuild-env

Edit repositories on GitHub

CancelOK

☐ I understand that GitHub content for the selected repositories will be transferred to this GCP project to provide the connected service. Principals with access to this GCP project with sufficient permissions will be able to create and run triggers on these repositories, based on transferred GitHub content. I also understand that content from all GitHub App triggers in this GCP project may be transferred to GitHub in order to provide functionality like showing trigger names in GitHub build results. This will apply to all existing and future GitHub App triggers in this project. [Learn more](#)

Connect

8:27 PM

13-Oct-2025

Create trigger – Cl...xGitHubxDevice Activationx+

console.cloud.google.com/cloud-build/triggers;region=u...Incognito (2)

Google Cloudqwiklabs-gcp-03-a83160aaeca9

Cloud Build / Triggers / Create trigger

Create trigger

Repository service

☒ Cloud Build repositories

☐ Developer Connect Preview

Developer Connect is a new service that provides a streamlined way for you to manage connections to source code management platforms outside of Google Cloud

Repository generation

☒ 1st gen

☐ 2nd gen

Repository *

BSSE23058/hello-cloudbuild-app (GitHub App)

Select the repository to watch for events and clone when the trigger is invoked

Branch *

.*

Trigger only for a branch that matches the given regular expression [Learn more](#)

☐ Invert Regex

Matches the branch: master

[Show included and ignored files filters](#)

Configuration

Type

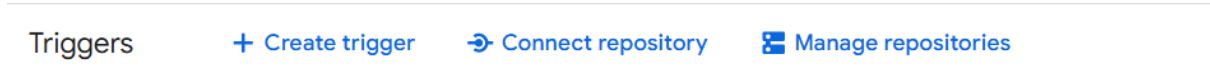
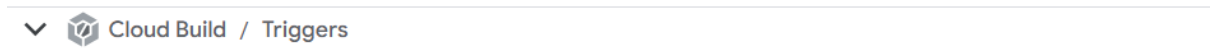
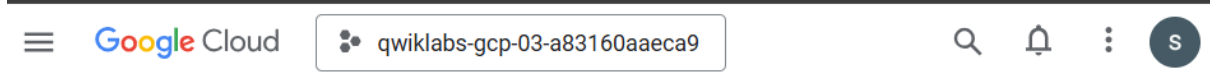
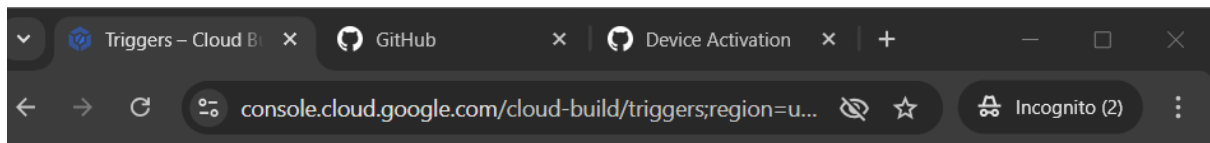
Connected repository

"BSSE23058/hello-cloudbuild-app"

X

8:27 PM

13-Oct-2025



A Cloud Build trigger automatically starts a build whenever you make any changes to your source code or some other incoming event.

Filter Enter property name or value						
Name ↑	Region	Description	Repository	Event	Run	
hello-cloudbuild	us-west1	—	BSSE23058/hello-cloudbuild-app	Push	Run	

'hello-cloudbuild' has been created on
BSSE23058/hello-cloudbuild-app



Create trigger – Cl xGitHub xDevice Activation x+ x

console.cloud.google.com/cloud-build/triggers/add;regi... Incognito (2)

Google Cloud

Cloud Build / Triggers

Create trigger

Repository generation

- 1st gen
- 2nd gen

Repository *

- Filter Type to filter
- BSSE23058/hello-cloud
- Invert Regex
- Show included and ignored

Configuration

- Type
 - Cloud Build configuration
 - Dockerfile
 - Buildpacks
- Location
 - Repository

Connect repository

Region: us-west1

1 Select source code management provider

2 Authenticate

3 Select repository

Select the GitHub repositories to connect to Cloud Build. Principals with access to this Google Cloud project will be able to create and run triggers on these repositories.

GitHub Account *

BSSE23058

Repository *

BSSE23058/hello-cloudbuild-env

☒

 I understand that GitHub content for the selected repositories will be transferred to this GCP project to provide the connected service. Principals with access to this GCP project with sufficient permissions will be able to create and run triggers on these repositories, based on transferred GitHub content. I also understand that content from all GitHub App triggers in this GCP project may be transferred to GitHub in order to provide functionality like showing trigger names in GitHub build results. This will apply to all existing and future GitHub App triggers in this project. [Learn more](#)

Connect



8:30 PM
13-Oct-2025

Triggers – Cloud B

GitHub

Device Activation

+

←

→

↺

console.cloud.google.com/cloud-build/triggers;region=u...

🔍

☆

🔒

Incognito (2)

⋮

☰

Google Cloud

🔍

🔔

⋮

S

☑

Cloud Build / Triggers

Triggers

+ Create trigger

➔ Connect repository

🗑 Manage repositories

A Cloud Build trigger automatically starts a build whenever you make any changes to your source code or some other incoming event.

☰ Filter

Enter property name or value

?

⋮

Name ↑	Region	Description	Repository	
hello-cloudbuild	us-west1	—	 BSSE23058/hello-cloudbuild-app	Run ⋮
hello-cloudbuild-deploy	us-west1	—	 BSSE23058/hello-cloudbuild-env	Run ⋮

'hello-cloudbuild-deploy' has been created on BSSE23058/hello-cloudbuild-env

×



8:31 PM

13-Oct-2025

Dashboard – Cloud xGitHub xDevice Activation x+

←→↻console.cloud.google.com/cloud-build/dashboard;region...Incognito (2)

Google Cloudqwiklabs-gcp-03-a83160aaeca9

Cloud Build / Dashboard

Dashboard

manually submitted builds, on the History page.

Filter Enter property name or value

Running: BSSE23058/hello-cloudbuild-app - hello-cloudbuild

Latest Build	Durati...	Trigger description	Source	Commit
10/13/25, 8:34...	00:00:35	—	...	1db9b...

Build History

[View all](#)

Average Duration ?

—

Pass - Fail % ?

—

Running: BSSE23058/hello-cloudbuild-env - hello-cloudbuild-deploy

Latest Build	Durati...	Trigger description	Source	Commit
10/13/25, 8:34...	00:00:20	—	...	010f63...

Build History

[View all](#)

Average Duration ?

—

Pass - Fail % ?

—

Build started on branch master for trigger hello-cloudbuild-deployShow X

8:34 PM13-Oct-2025

Cloud Build - qwiklabs-gcp-03

Google Cloud

qwiklabs-gcp-03-a83160aaeca9

Search (/) for resources, docs, products, and more

Cloud Build / Dashboard

Dashboard

manually submitted builds, on the History page.

Filter Enter property name or value

Successful: BSSE23058/hello-cloudbuild-app - hello-cloudbuild

Latest Build 10/13/25, 8:34 Durati... 00:00:38 Trigger description Source Commit 1db9b...

Build History View all

Average Duration 00:00:38

Pass - Fail % 100% - 0%

Failed: BSSE23058/hello-cloudbuild-env - hello-cloudbuild-deploy

Latest Build 10/13/25, 8:47 Durati... 00:00:22 Trigger description Source Commit 010f63 Failed step 1. Build

Build History View all

Average Duration 00:00:18

Pass - Fail % 0% - 100%

Recommended for you

- Build configuration overview
- Quickstart: Build
- Creating a basic build configuration file
- Creating and managing build triggers
- Cloud builders

22°C Clear

Search

Software Development & Con... Implement CI/CD Pipelines on GoogleCloudSkillsboost/Goog... Google Gemini

cloudskillsboost.google/paths/76/course_templates/691/labs/594612#step8

Google Cloud

Dashboard Explore Paths Subscriptions

Apply your skills in Google Cloud console Get started

DevSecOps > Course > Implement CI/CD Pipelines on Google Cloud >

10 Implement CI/CD Pipelines on Google Cloud

Course - 1 Complete hour 15 minutes

Course overview

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

End Lab 00:54:23

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username student-00-21b18e955805i

Password 36EUq9V9TSP

Project ID qwiklabs-gcp-03-a83160aaeca9

Previous

```
# unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

# [START cloudbuild]
steps:
# This step runs the unit tests on the app
- name: 'python:3.7-slim'
  id: Test
  entrypoint: /bin/sh
  args:
    - -c
    - 'pip install flask && python test_app.py -v'

# This step builds the container image.
- name: 'gcr.io/cloud-builders/docker'
  id: Build
  args:
    - 'build'
```

Lab instructions and tasks 85/100

GSP1077

Overview

Objectives

Setup and requirements

Task 1. Initialize your lab

Task 2. Create the Git repositories in GitHub repositories

Task 3. Create a container image with Cloud Build

Task 4. Create the Continuous Integration

Next

Software Development & Con...Implement CI/CD Pipelines on...GoogleCloudSkillsboost/Googl...Google Gemini

cloudskillsboost.google/paths/76/course_templates/691/labs/594612#step8

Google CloudDashboardExplorePathsSubscriptions

Apply your skills in Google Cloud consoleGet started

10Implement CI/CD Pipelines on Google Cloud

Course - 1Completehour 15 minutes

Course overview

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

End Lab00:54:09

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Usernamestudent-00-21b18e955805i

Password36EUq9V9TSvp

Project IDqw1klabs-gcp-03-a83160ai

Previous

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[START cloudbuild]

steps:

- name: 'python:3.7-slim'

id: Test

entrypoint: /bin/sh

args:

- -c

- 'pip install flask && python test_app.py -v'

This step builds the container image.

- name: 'gcr.io/cloud-builders/docker'

id: Build

args:

- 'build'

Build

Create the Continuous Integration (CI) PipelineCheck my progress20 / 20

Accessing GitHub from a build via SSH keysCheck my progress20 / 20

Create the Test Environment and CD PipelineCheck my progress5 / 20

Task 2. Create the Git repositories in GitHub repositories

Task 3. Create a container image with Cloud Build

Task 4. Create the Continuous Integration

Next

22°C Clear

Search

8:47 PM13-Oct-2025

3. Continuous Delivery with Google Cloud Deploy

Dashboard – qwiklabs-gcp-04-93e61e9975d5 x +

← → ↻ console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5 Incognito

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor

```
emini CLI.
Your Cloud Platform project in this session is set to qwiklabs-gcp-04-93e61e9975d5.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5) $ export PROJECT_ID=$(gcloud config get-value project)
export REGION=us-east1
gcloud config set compute/region $REGION
Your active configuration is: [cloudshell-17993]
Updated property [compute/region].
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5) $
```

The image shows a web browser window displaying the Google Cloud console dashboard for the project 'qwiklabs-gcp-04-93e61e9975d5'. Below the dashboard, a terminal window is open, showing the execution of the Gemini CLI. The terminal output indicates that the Cloud Platform project is set to 'qwiklabs-gcp-04-93e61e9975d5' and the region is 'us-east1'. It also shows the successful completion of the 'gcloud services enable container.googleapis.com' command.

```
(qwiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor
emini CLI.
Your Cloud Platform project in this session is set to quiklabs-gcp-04-93e61e9975d5.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$ export PROJECT_ID=$(gcl
oud config get-value project)
export REGION=us-east1
gcloud config set compute/region $REGION
Your active configuration is: [cloudshell-17993]
Updated property [compute/region].
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$ gcloud services enable
\
container.googleapis.com \
clouddeploy.googleapis.com
Operation "operations/acat.p2-143823944593-88ba2002-e923-4e65-9ed8-ea23392b35f1" finished su
ccessfully.
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$
```

Dashboard - qwiklabs-gcp-04-... x +

console.cloud.google.com/home/dashboard?project=qwikl... Incognito

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor

```
clouddeploy.googleapis.com
Operation "operations/acat.p2-143823944593-88ba2002-e923-4e65-9ed8-ea23392b35f1" finished successfully.
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$ gcloud container clusters create test --node-locations=us-east1-d --num-nodes=1 --async
gcloud container clusters create staging --node-locations=us-east1-d --num-nodes=1 --async
gcloud container clusters create prod --node-locations=us-east1-d --num-nodes=1 --async
Note: Your Pod address range ('--cluster-ipv4-cidr') can accommodate at most 1008 node(s).
NAME: test
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
Note: Your Pod address range ('--cluster-ipv4-cidr') can accommodate at most 1008 node(s).
NAME: staging
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
Note: Your Pod address range ('--cluster-ipv4-cidr') can accommodate at most 1008 node(s).
NAME: prod
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$
```

Dashboard - qwiklabs-gcp-04-... x +

console.cloud.google.com/home/dashboard?project=qwikl... Incognito

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor

```
gcloud container clusters create prod --node-locations=us-east1-d --num-nodes=1 --async
Note: Your Pod address range (--cluster-ipv4-cidr) can accommodate at most 1008 node(s).
NAME: test
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
Note: Your Pod address range (--cluster-ipv4-cidr) can accommodate at most 1008 node(s).
NAME: staging
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
Note: Your Pod address range (--cluster-ipv4-cidr) can accommodate at most 1008 node(s).
NAME: prod
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$ gcloud container clusters list --format="csv(name,status)"
name,status
prod,PROVISIONING
staging,PROVISIONING
test,PROVISIONING
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$

student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$ gcloud services enable artifactregistry.googleapis.com
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$ gcloud artifacts repositories create web-app \
--description="Image registry for tutorial web app" \
--repository-format=docker \
--location=$REGION
Create request issued for: [web-app]
Waiting for operation [projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/operations/500a5e39-c817-4457-a266-669bef64b810] to complete...done.
Created repository [web-app].
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$
```

Dashboard - qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor

```
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ gcloud services enable cloudbuild.googleapis.com
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ gsutil mb -p $PROJECT_ID gs://$PROJECT_ID/_cloudbuild
Creating gs://qwiklabs-gcp-04-93e61e9975d5_cloudbuild/...
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ cd web
skaffold build --interactive=false \
--default-repo $REGION-docker.pkg.dev/$PROJECT_ID/web-app \
--file-output artifacts.json
cd ..
Generating tags...
- leeroy-web -> us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-web:c3cae80
- leeroy-app -> us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-app:c3cae80
Checking cache...
- leeroy-web: Not found. Building
- leeroy-app: Not found. Building
Starting build...
Building 2 artifacts in parallel
Building [leeroy-web]...
Pushing code to gs://qwiklabs-gcp-04-93e61e9975d5_cloudbuild/source/qwiklabs-gcp-04-93e61e9975d5-2da244b0-41b5-48b2-bb68-14115efecf20.tar.gz
Logs are available at
https://storage.cloud.google.com/qwiklabs-gcp-04-93e61e9975d5_cloudbuild/log-a08d2fe3-b6b4-4f00-8382-23f07a215c2d.txt
starting build "a08d2fe3-b6b4-4f00-8382-23f07a215c2d"

FETCHSOURCE
Fetching storage object: gs://qwiklabs-gcp-04-93e61e9975d5_cloudbuild/source/qwiklabs-gcp-04-93e61e9975d5-2da244b0-41b5-48b2-bb68-14115efecf20.tar.gz#1760363028234840
Copying gs://qwiklabs-gcp-04-93e61e9975d5_cloudbuild/source/qwiklabs-gcp-04-93e61e9975d5-2da244b0-41b5-48b2-bb68-14115efecf20.tar.gz#1760363028234840...
/ [1 files][ 981.0 B/ 981.0 B]
```

6:45 PM 13-Oct-2025

Dashboard - qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) Gemini CLI Editor

```
Step 1/7 : FROM golang:1.12.9-alpine3.10 as builder
1.12.9-alpine3.10: Pulling from library/golang
9d48c3bd43c5: Pulling fs layer
7f94eaf8af20: Pulling fs layer
9fe9984849c1: Pulling fs layer
cf0db633a67d: Pulling fs layer
0f7136d71739: Pulling fs layer
cf0db633a67d: Waiting
0f7136d71739: Waiting
9fe9984849c1: Verifying Checksum
9fe9984849c1: Download complete
9d48c3bd43c5: Verifying Checksum
9d48c3bd43c5: Download complete
7f94eaf8af20: Verifying Checksum
7f94eaf8af20: Download complete
9d48c3bd43c5: Pull complete
0f7136d71739: Verifying Checksum
0f7136d71739: Download complete
7f94eaf8af20: Pull complete
9fe9984849c1: Pull complete
cf0db633a67d: Verifying Checksum
cf0db633a67d: Download complete
cf0db633a67d: Pull complete
0f7136d71739: Pull complete
Digest: sha256:e0660b4f1e68e0d408420acb874b396fc6dd25e7c1d03ad36e7d6d1155a4dff6
Status: Downloaded newer image for golang:1.12.9-alpine3.10
---> e0d646523991
Step 2/7 : COPY web.go .
---> cfe62aa9cad5
Step 3/7 : RUN go build -o /web .
---> Running in 4f9705221831
Removing intermediate container 4f9705221831
---> 2927794ee11b
Step 4/7 : FROM alpine:3.10
3.10: Pulling from library/alpine
```

6:45 PM 13-Oct-2025

Dashboard - qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) Gemini CLI Editor

```
package: us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-web
tags:
- c3cae80
updateTime: '2025-10-13T13:44:12.373743Z'
version: sha256:13b9834adf669cb8b1345f0fa9470926d99e39a39de2cfc1af2aaba426d2737e
---
createTime: '2025-10-13T13:44:27.433055Z'
metadata:
  buildTime: null
  imageSizeBytes: '5411'
  mediaType: application/vnd.oci.image.manifest.v1+json
  name: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/repositories/web-app/docker
Images/leeroy-web@sha256:dd3821ea572f0f7e161bcl3a3288182c1b35f4662d9b0a94466eee16d66a777
package: us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-web
tags: []
updateTime: '2025-10-13T13:44:27.433055Z'
version: sha256:dd3821ea572f0f7e161bcl3a3288182c1b35f4662d9b0a94466eee16d66a777
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $ cat web/artifacts.json | jq
{
  "builds": [
    {
      "imageName": "leeroy-web",
      "tag": "us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-web:c3cae80@sha256:13b9834adf669cb8b1345f0fa9470926d99e39a39de2cfc1af2aaba426d2737e"
    },
    {
      "imageName": "leeroy-app",
      "tag": "us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-app:c3cae80@sha256:a3fe1f6cf762db0c9e04378fe5bdb4f2270eabcf11163e1fd92b09561943afbd"
    }
  ]
}
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $
```

6:45 PM 13-Oct-2025

```
93e61e9975d5) $ gcloud config set deploy/region $REGION
cp clouddeploy-config/delivery-pipeline.yaml.template clouddeploy-config/delivery-pipeline.yaml
Updated property [deploy/region].
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $ gcloud container clusters list --format="csv(name,status)"
name,status
prod,RUNNING
staging,RUNNING
test,RUNNING
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $
```

6:46 PM 13-Oct-2025

Dashboard - qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5&pli=1&cloudshell=true

Google Cloud qwiklabs-gcp-04-93e61e9975d5 Search (/) for resources, docs, products, and more

Dashboard Activity Recommendations Customize

CLOUD SHELL Terminal (qwiklabs-gcp-04-93e61e9975d5) Gemini CLI + Open Editor

```
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ CONTEXTS=("test" "staging" "prod")
for CONTEXT in $(CONTEXTS[@]);
do
  gcloud container clusters get-credentials $(CONTEXT) --region $(REGION)
  kubectl config rename-context gke_${PROJECT_ID}_${REGION}_${CONTEXT} $(CONTEXT)
done
Fetching cluster endpoint and auth data.
kubeconfig entry generated for test.
Context "gke_qwiklabs-gcp-04-93e61e9975d5_us-east1_test" renamed to "test".
Fetching cluster endpoint and auth data.
kubeconfig entry generated for staging.
Context "gke_qwiklabs-gcp-04-93e61e9975d5_us-east1_staging" renamed to "staging".
Fetching cluster endpoint and auth data.
kubeconfig entry generated for prod.
Context "gke_qwiklabs-gcp-04-93e61e9975d5_us-east1_prod" renamed to "prod".
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ for CONTEXT in $(CONTEXTS[@])
do
  kubectl --context $(CONTEXT) apply -f kubernetes-config/web-app-namespace.yaml
done
namespace/web-app created
namespace/web-app created
namespace/web-app created
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ for CONTEXT in $(CONTEXTS[@])
do
  envsubst < clouddeploy-config/target-SCONTEXT.yaml.template > clouddeploy-config/target-SCONTEXT.yaml
  gcloud beta deploy apply --file clouddeploy-config/target-SCONTEXT.yaml
done
Waiting for the operation on resource projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/test...done.
Created Cloud Deploy resource: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/test.
Waiting for the operation on resource projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/staging...done.
Created Cloud Deploy resource: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/staging.
Waiting for the operation on resource projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/prod...done.
Created Cloud Deploy resource: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/prod.
```

student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)\$ cat clouddeploy-config/target-test.yaml

```
# Copyright 2021 Google LLC
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the license is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

apiVersion: deploy.cloud.google.com/v1beta1
kind: Target
metadata:
  name: test
  description: test cluster
gke:
  cluster: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/clusters/test
```

student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)\$ cat clouddeploy-config/target-prod.yaml

```
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#
# http://www.apache.org/licenses/LICENSE-2.0
#
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# distributed under the license is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
```

Dashboard - qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5&pli=1&cloudshell=true

Google Cloud qwiklabs-gcp-04-93e61e9975d5 Search (/) for resources, docs, products, and more

Dashboard Activity Recommendations Customize

CLOUD SHELL Terminal (qwiklabs-gcp-04-93e61e9975d5) Gemini CLI + Open Editor

```
- RENDER
- DEPLOY
- VERIFY
- PREDEPLOY
- POSTDEPLOY

gke:
  cluster: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/clusters/test
  name: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/test
  targetId: test
  uid: 4b75220d3dc24a5d976c2fed1998d8a7
  updateTime: '2025-10-13T13:47:49.138946779z'
  createTime: '2025-10-13T13:48:00.035041605z'
  description: prod cluster
  etag: fa2c44801ec10792
  executionConfig:
    artifactStorage: gs://us-east1.deploy-artifacts.qwiklabs-gcp-04-93e61e9975d5.appspot.com
    defaultPool:
      artifactStorage: gs://us-east1.deploy-artifacts.qwiklabs-gcp-04-93e61e9975d5.appspot.com
      serviceAccount: 143823944593-compute@developer.gserviceaccount.com
      executionTimeout: 3600s
      serviceAccount: 143823944593-compute@developer.gserviceaccount.com
    usages:
      - RENDER
      - DEPLOY
      - VERIFY
      - PREDEPLOY
      - POSTDEPLOY
  gke:
    cluster: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/clusters/prod
    name: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/prod
    requireApproval: true
    targetId: prod
    uid: bdecc8deb6884a838e91b10e876b099a
student_04_ead334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $
```

24°C Clear 6:49 PM 13-Oct-2025

Dashboard - qwiklabs-gcp-04-... x +

console.cloud.google.com/home/dashboard?project=qwikl... Incognito

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(quiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor

```
id: stable
state: IN_PROGRESS
state: IN_PROGRESS
targetId: staging
uid: 223b1c779b13422390c6f3c15b291077
---
approvalState: DOES_NOT_NEED_APPROVAL
createTime: '2025-10-13T13:52:55.316741Z'
deployEndTime: '2025-10-13T13:53:35.041573Z'
deployStartTime: '2025-10-13T13:53:08.573794551Z'
deployingBuild: projects/143823944593/locations/us-east1/builds/97d5c4ae-3c58-42f6-a808-41efb1120ba6
enqueueTime: '2025-10-13T13:53:08.222239Z'
etag: 17feab07705c0b59
name: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/deliveryPipelines/web-app/releases/web-app-001/rollouts/web-app-001-to-test-0001
phases:
- deploymentJobs:
  deployJob:
    deployJob: {}
    id: deploy
    jobRun: projects/143823944593/locations/us-east1/deliveryPipelines/web-app/releases/web-app-001/rollouts/web-app-001-to-test-0001/jobRuns/b967eb05-f3e1-4806-85c1-344f37b559f7
    state: SUCCEEDED
    verifyJob:
      id: verify
      state: DISABLED
      verifyJob: {}
  id: stable
  state: SUCCEEDED
state: SUCCEEDED
targetId: test
uid: f657cf4351e84b6b97396c6a6dcea6c7
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (quiklabs-gcp-04-93e61e9975d5) $
```

6:54 PM 13 Oct 2025

Dashboard - qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) Gemini CLI Editor

```
approvalState: DOES_NOT_NEED_APPROVAL
createTime: '2025-10-13T13:52:55.316741Z'
deployEndTime: '2025-10-13T13:53:35.041573Z'
deployStartTime: '2025-10-13T13:53:08.573794551Z'
deployingBuild: projects/143823944593/locations/us-east1/builds/97d5c4ae-3c58-42f6-a808-41efb1120ba6
enqueueTime: '2025-10-13T13:53:08.222239Z'
etag: 17feab07705c0b59
name: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/deliveryPipelines/web-app/releases/web-app-001/rollouts/web-app-001-to-test-0001
phases:
- deploymentJobs:
  deploymentJob:
    deploymentJob: {}
    id: deploy
    jobRun: projects/143823944593/locations/us-east1/deliveryPipelines/web-app/releases/web-app-001/rollouts/web-app-001-to-test-0001/jobRuns/b967eb05-f3e1-4806-85c1-344f37b559f7
    state: SUCCEEDED
    verifyJob:
      id: verify
      state: DISABLED
      verifyJob: {}
    id: stable
    state: SUCCEEDED
state: SUCCEEDED
targetId: test
uid: f657cf4351e84b6b97396c6a6dcea6c7
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $ kubectlx prod
kubectl get all -n web-app
Switched to context "prod".
NAME                                DESIRED  CURRENT  READY  AGE
replicaset.apps/leeroy-web-656d5c79ff  1         0         0      1s
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $
```

6:55 PM 13-Oct-2025

Software Development & Con...

Implement CI/CD Pipelines on

New Tab

cloudskillsboost.google/paths/76/course_templates/691/labs/594613

Google Cloud

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Apply your skills in Google Cloud console

Get started

10

Implement CI/CD Pipelines on Google Cloud

Course - 1 hour < 1% 15 minutes complete

Course overview

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

End Lab

01:07:15

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Project ID

qwiklabs-gcp-04-93e61e9f

Username

student-04-6ad334ed764fi

Password

ScddhQMyIF5A

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Check complete. Points earned: 10. Message: Assessment Completed!

Next

Lab instructions and tasks

100/100

GSP1079

Overview

Objectives

Setup and requirements

Task 1. Set the variables

Task 2. Create three GKE clusters

Task 3. Prepare the web application container image

Task 4. Build and deploy the container images to

Implement CI/CD

24°C Clear

Search

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Implement CI/CD Pipelines on Google Cloud

Course - 1 hour < 1% 15 minutes complete

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Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

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01:07:05

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Username

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Password

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Checkpoints

Create three GKE clusters

Check my progress

10 / 10

Create the web-app repository

Check my progress

10 / 10

Build and deploy the container images

Check my progress

10 / 10

Task 2. Create three GKE clusters

Task 3. Prepare the web application container image

Task 4. Build and deploy the container images to

Next

Implement CI/CD

24°C Clear

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10Implement CI/CD Pipelines on Google Cloud

Course · 1 hour · < 1% complete15 minutes

Course overview

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

End Lab01:06:52

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Usernamestudent-04-6ad334ed764fi

PasswordScddhQMyIFSA

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Create the delivery pipelineCheck my progress10 / 10

Configure the deployment targetsCheck my progress20 / 20

Create a releaseCheck my progress20 / 20

Promote the application to stagingCheck my progress10 / 10

Task 2. Create three GKE clusters

Task 3. Prepare the web application container image

Task 4. Build and deploy the container images to

Next

24°C Clear

Search

6:56 PM13-Oct-2025

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Google CloudDashboardExplorePathsSubscriptions

Apply your skills in Google Cloud consoleGet started

10Implement CI/CD Pipelines on Google Cloud

Course · 1 hour · < 1% complete15 minutes

Course overview

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

End Lab01:06:43

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Open Google Cloud console

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Usernamestudent-04-6ad334ed764fi

PasswordScddhQMyIFSA

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Create a releaseCheck my progress20 / 20

Promote the application to stagingCheck my progress10 / 10

Promote the application to prodCheck my progress10 / 10

Task 2. Create three GKE clusters

Task 3. Prepare the web application container image

Task 4. Build and deploy the container images to

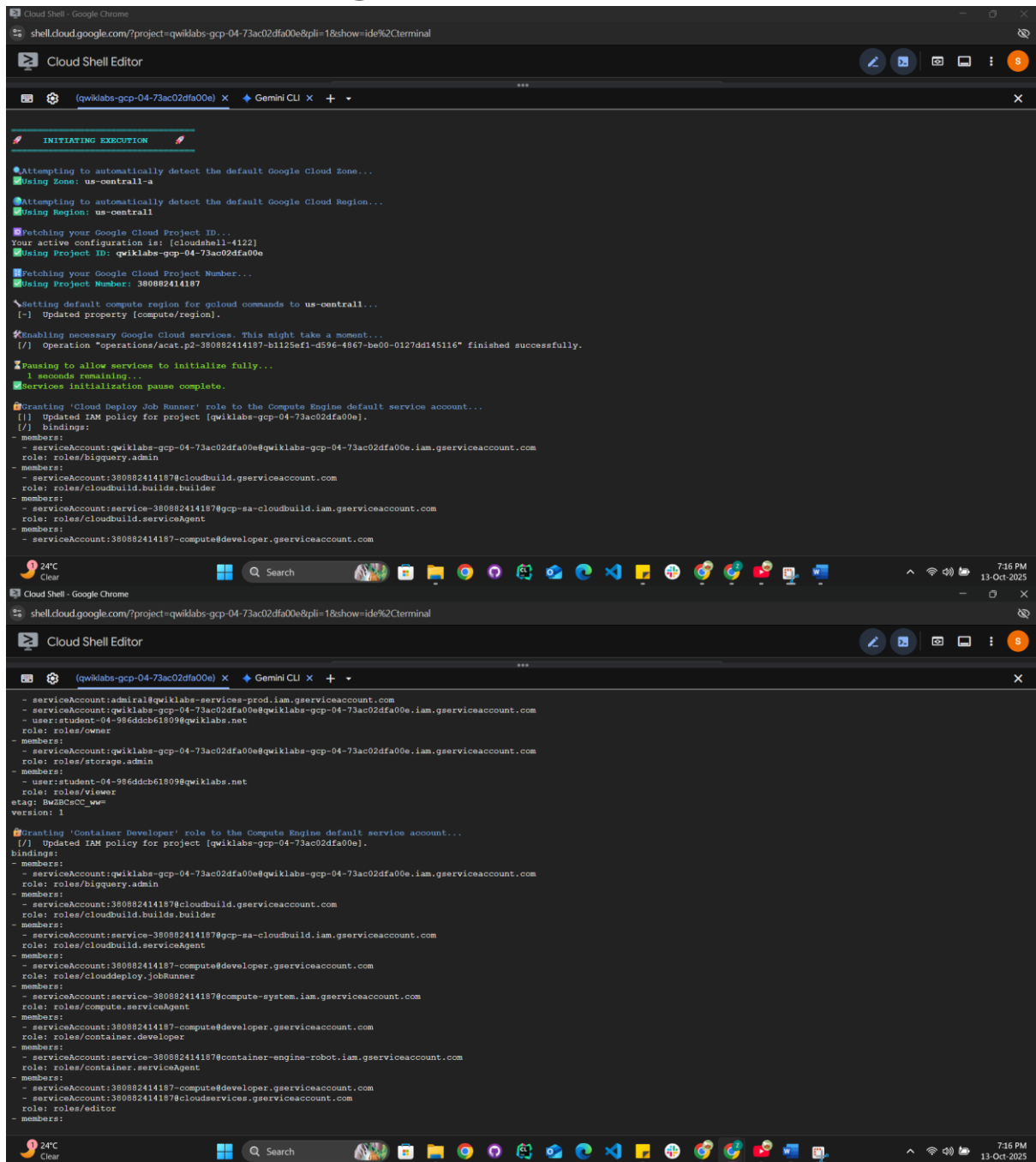
Next

24°C Clear

Search

6:56 PM13-Oct-2025

4. Implement CI CD Pipelines on Google Cloud Challenge Lab



```
Cloud Shell - Google Chrome
shell.cloud.google.com/project-qwiklabs-gcp-04-73ac02dfa00e&pli=1&show=ide%2Cterminal

Cloud Shell Editor

(qwiklabs-gcp-04-73ac02dfa00e) x Gemini CLI x +

INITIATING EXECUTION

Attempting to automatically detect the default Google Cloud Zone...
Using Zone: us-central1-a

Attempting to automatically detect the default Google Cloud Region...
Using Region: us-central1

Fetching your Google Cloud Project ID...
Your active configuration is: [cloudshell-4122]
Using Project ID: qwiklabs-gcp-04-73ac02dfa00e

Fetching your Google Cloud Project Number...
Using Project Number: 380882414187

Setting default compute region for gcloud commands to us-central1...
[-] Updated property [compute/region].

Enabling necessary Google Cloud services. This might take a moment...
[!] Operation "operations/acet.p2-380882414187-b1125ef1-d596-4667-be00-0127dd145116" finished successfully.

Pausing to allow services to initialize fully...
1 seconds remaining...
Services initialization pause complete.

Granting 'Cloud Deploy Job Runner' role to the Compute Engine default service account...
[!] Updated IAM policy for project [qwiklabs-gcp-04-73ac02dfa00e].
bindings:
- members:
  - serviceAccount:qwiklabs-gcp-04-73ac02dfa00e@qwiklabs-gcp-04-73ac02dfa00e.iam.gserviceaccount.com
  role: roles/bigquery.admin
- members:
  - serviceAccount:380882414187@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-380882414187@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:380882414187-compute@developer.gserviceaccount.com

24°C Clear
Search
Cloud Shell - Google Chrome
shell.cloud.google.com/project-qwiklabs-gcp-04-73ac02dfa00e&pli=1&show=ide%2Cterminal

Cloud Shell Editor

(qwiklabs-gcp-04-73ac02dfa00e) x Gemini CLI x +

- serviceAccount:admiral@qwiklabs-services-prod.iam.gserviceaccount.com
- serviceAccount:qwiklabs-gcp-04-73ac02dfa00e@qwiklabs-gcp-04-73ac02dfa00e.iam.gserviceaccount.com
- user:student-04-986ddcb61809@qwiklabs.net
  role: roles/owner
- members:
  - serviceAccount:qwiklabs-gcp-04-73ac02dfa00e@qwiklabs-gcp-04-73ac02dfa00e.iam.gserviceaccount.com
  role: roles/storage.admin
- members:
  - user:student-04-986ddcb61809@qwiklabs.net
  role: roles/viewer
etag: Bw2BCaCC_wm
version: 1

Granting 'Container Developer' role to the Compute Engine default service account...
[!] Updated IAM policy for project [qwiklabs-gcp-04-73ac02dfa00e].
bindings:
- members:
  - serviceAccount:qwiklabs-gcp-04-73ac02dfa00e@qwiklabs-gcp-04-73ac02dfa00e.iam.gserviceaccount.com
  role: roles/bigquery.admin
- members:
  - serviceAccount:380882414187@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-380882414187@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:380882414187-compute@developer.gserviceaccount.com
  role: roles/clouddeploy.jobRunner
- members:
  - serviceAccount:service-380882414187@compute-system.iam.gserviceaccount.com
  role: roles/compute.serviceAgent
- members:
  - serviceAccount:380882414187-compute@developer.gserviceaccount.com
  role: roles/container.developer
- members:
  - serviceAccount:service-380882414187@container-engine-robot.iam.gserviceaccount.com
  role: roles/container.serviceAgent
- members:
  - serviceAccount:380882414187-compute@developer.gserviceaccount.com
  - serviceAccount:380882414187@cloudservices.gserviceaccount.com
  role: roles/editor
- members:
```

```
Cloud Shell - Google Chrome
shell.cloud.google.com/?project=qwiklabs-gcp-04-73ac02dfa00e&pli=1&show=ide%2Cterminal

Cloud Shell Editor

(qwiklabs-gcp-04-73ac02dfa00e) x Gemini CLI x +

Creating Artifact Registry repository 'cicd-challenge' for Docker images in us-central1...
[!] Note: Create request issued for: [cicd-challenge]
Waiting for operation [projects/qwiklabs-gcp-04-73ac02dfa00e/locations/us-central1/operations/6619079e-cde4-498f-b082-80d7aff09b3f] to complete...
[!] .....done.
[-] Created repository [cicd-challenge].

Creating GKE cluster 'cd-staging' in zone us-central1-a (asynchronously)...
[!] Note: Your Pod address range ('--cluster-ip4-cidr') can accommodate at most 1008 node(s).
[!] NAME: cd-staging
TYPE:
LOCATION: us-central1
TARGET:
STATUS MESSAGE:
STATUS: PROVISIONING
START TIME:
END TIME:

Creating GKE cluster 'cd-production' in zone us-central1-a (asynchronously)...
[!] Note: Your Pod address range ('--cluster-ip4-cidr') can accommodate at most 1008 node(s).
[!] NAME: cd-production
TYPE:
LOCATION: us-central1
TARGET:
STATUS MESSAGE:
STATUS: PROVISIONING
START TIME:
END TIME:

Navigating to home directory...
Cloning 'cloud-deploy-tutorials' repository from GitHub...
Cloning into 'cloud-deploy-tutorials'...
[!] remote: Enumerating objects: 1620, done.
remote: Counting objects: 100% (151/151), done.
remote: Compressing objects: 100% (85/85), done.
remote: Total 1620 (delta 103), reused 96 (delta 96), pack-reused 1469 (from 1)
Receiving objects: 100% (1620/1620), 588.13 KiB | 2.24 MiB/s, done.
Resolving deltas: 100% (795/795), done.
Changing directory to 'cloud-deploy-tutorials'...
Checking out a specific commit (c3cae80) silently...
Changing directory to 'tutorials/base'...

Generating Scaffold configuration (scaffold.yaml) from template...
Updating Scaffold configuration with Project ID: qwiklabs-gcp-04-73ac02dfa00e...

24°C Clear
Cloud Shell - Google Chrome
shell.cloud.google.com/?project=qwiklabs-gcp-04-73ac02dfa00e&pli=1&show=ide%2Cterminal

Cloud Shell Editor

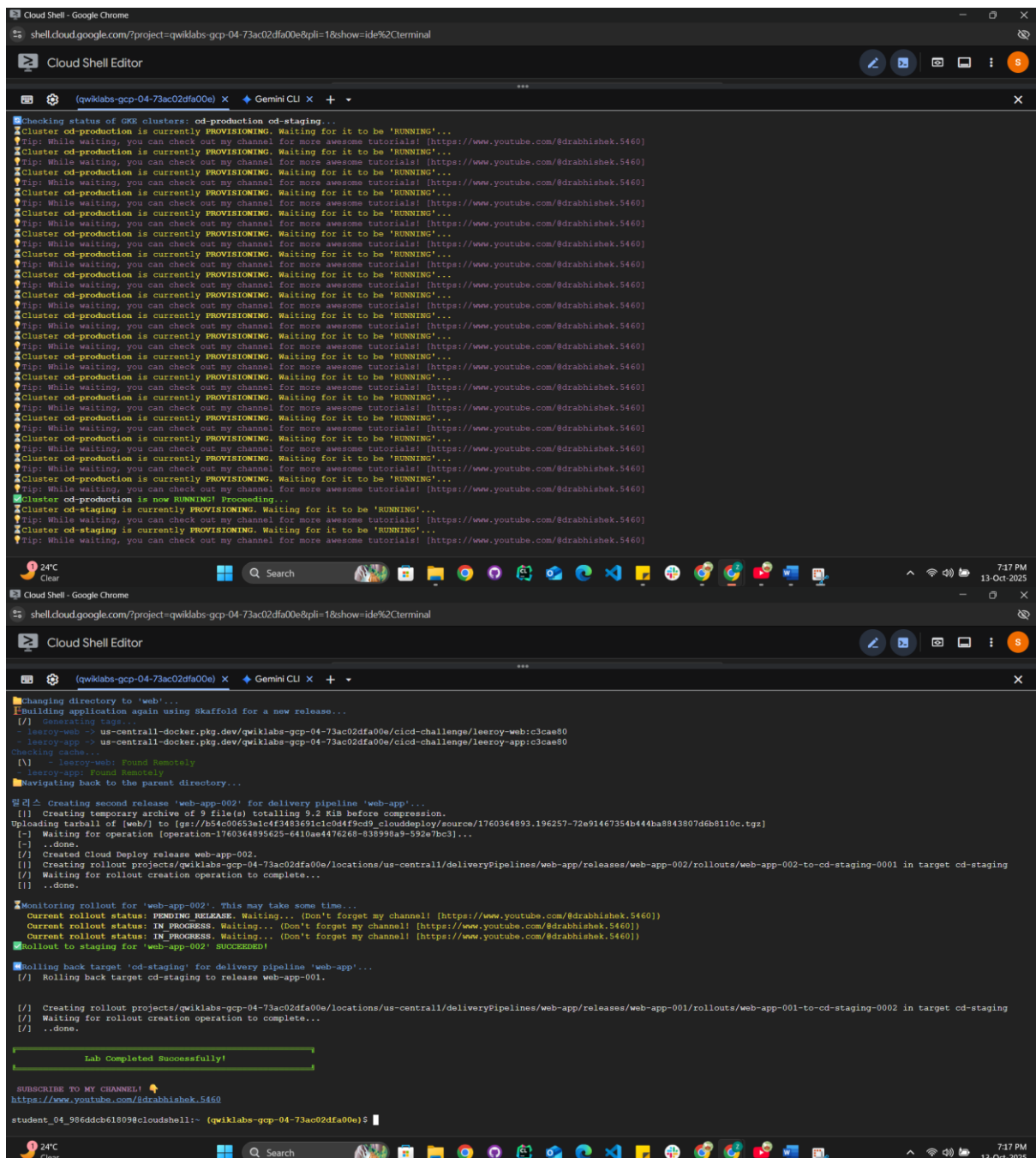
(qwiklabs-gcp-04-73ac02dfa00e) x Gemini CLI x +

Bucket not found. Creating bucket in region us-central1. Open a new tab with project...
[!] Creating gs://qwiklabs-gcp-04-73ac02dfa00e_cloudbuild/...

Changing directory to 'web'...
Building application using Scaffold and outputting artifacts to 'artifacts.json'...
Repository: us-central1-docker.pkg.dev/qwiklabs-gcp-04-73ac02dfa00e/cicd-challenge
[!] Generating logs...
- leeroyw [-] us-central1-docker.pkg.dev/qwiklabs-gcp-04-73ac02dfa00e/cicd-challenge/leeroy-web:c3cae80
- leeroy-app -> us-central1-docker.pkg.dev/qwiklabs-gcp-04-73ac02dfa00e/cicd-challenge/leeroy-app:c3cae80
Checking cache...
[!] - leeroy-web: Not found. Building
- leeroy-app: Not found. Building
Starting build...
Building 2 artifacts in parallel
Building [leeroy-web]...
[!] Pushing code to gs://qwiklabs-gcp-04-73ac02dfa00e_cloudbuild/source/qwiklabs-gcp-04-73ac02dfa00e-e340af6f-858d-4976-a940-105934aa0128.tar.gz
[!] Logs are available at
https://source.cloud.google.com/qwiklabs-gcp-04-73ac02dfa00e_cloudbuild/log-7527769f-0f90-40d0-9884-2a3f1e6ba2d.txt
[!] starting build "7527769f-0f90-40d0-9884-2a3f1e6ba2d"

FETCHSOURCE
Fetching storage object: gs://qwiklabs-gcp-04-73ac02dfa00e_cloudbuild/source/qwiklabs-gcp-04-73ac02dfa00e-e340af6f-858d-4976-a940-105934aa0128.tar.gz#1760364368920099
[!] Copying gs://qwiklabs-gcp-04-73ac02dfa00e_cloudbuild/source/qwiklabs-gcp-04-73ac02dfa00e-e340af6f-858d-4976-a940-105934aa0128.tar.gz#1760364368920099...
/ [1 files] [ 986.0 B / 986.0 B ]
Operation completed over 1 objects/986.0 B.

BUILD
Already have image (with digest): gcr.io/cloud-builders/docker
Sending build context to Docker daemon 4.096kB
Step 1/7 : FROM golang:1.12.9-alpine3.10 as builder
[!] 1.12.9-alpine3.10: Pulling from library/golang
9d48c3bd43c5: Pulling fs layer
7f94eaf8af20: Pulling fs layer
9fe9984849c1: Pulling fs layer
cf0db633a67d: Pulling fs layer
0f7136d71739: Pulling fs layer
cf0db633a67d: Waiting
0f7136d71739: Waiting
7f94eaf8af20: Verifying Checksum
7f94eaf8af20: Download complete
9d48c3bd43c5: Verifying Checksum
9d48c3bd43c5: Download complete
9fe9984849c1: Verifying Checksum
```



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10Implement CI/CD Pipelines on Google Cloud

Course · 1 hour 25%15 minutes complete

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

Implement CI/CD Pipelines on Google Cloud: Challenge Lab

End Lab01:12:28

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Open Google Cloud console

Project IDqwiklabs-gcp-04-73ac82d

Usernamestudent-04-986ddcb61889i

Passwordxmv4W9WCd4Xf

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Lab instructions and tasks100/100

GSP393

Overview

Setup

Challenge Scenario

Task 1. Prework - Set up environment, enable APIs and create clusters

Task 2. Build the images and upload to the repository

Task 3. Create the Delivery Pipeline

Next

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Working with Artifact Registry

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Implement CI/CD Pipelines on Google Cloud: Challenge Lab

End Lab01:12:15

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Project IDqwiklabs-gcp-04-73ac82d

Usernamestudent-04-986ddcb61889i

Passwordxmv4W9WCd4Xf

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Checkpoints

Validate that the container was created and added to the repositoryCheck my progress20 / 20

Verify that the Cloud Deploy targets have been createdCheck my progress20 / 20

Verify the release to the StagingCheck my progress20 / 20

Task 1. Prework - Set up environment, enable APIs and create clusters

Task 2. Build the images and upload to the repository

Task 3. Create the Delivery Pipeline

Next

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End Lab01:12:04

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Project ID

qw1k1abs-gcp-04-73ac02d

Username

student-04-986ddcb61809i

Password

xmv4IW9WCd4Xf

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Verify the release to the Staging environment

Check my progress

20 / 20

Verify the release to the Production environment

Check my progress

20 / 20

Verify the rollback ran successfully

Check my progress

20 / 20

environment, enable APIs and create clusters

Task 2. Build the images and upload to the repository

Task 3. Create the Delivery Pipeline

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